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Research on the application of CORDIC algorithm in the field of space-borne on-board signal processing

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Abstract

This paper briefly introduces the basic principle of the CORDIC algorithm, gives the specific design and implementation method of the general CORDIC soft core in the project based on the background application of Space-borne payload processing. The actual application shows that the design method can be used to process the signals that the conventional design method needs a lot of resources to achieve under the more demanding requirements of the hardware and software platform Application scenario.

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Keywords: CORDIC; FPGA; Spcace-borne; On-board Payload Processing

1. Overview

In recent years, with the rapid development of digital signal processing and digital integrated circuit technology, the function of satellite platform is becoming more and more complex. The real-time signal processing technology on satellite, represented by the functions of autonomous mission planning, on orbit target identification, real-time information distribution [1-2], is widely used. Due to the particularity of the satellite working environment, the resources of the signal processing system on the satellite are greatly restricted. The specific performance is that there is no protection of the earth's magnetic field in the satellite environment, the electronic products are radiated by space electromagnetic radiation, and the space single particle bombardment environment, which leads to the traditional high-speed signal processing equipment such as DSP, CUDA and so on can not be directly applied to the satellite environment [3]. At the same time, due to the particularity of the satellite working environment, there is

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little possibility of on orbit repair after its launch, which requires the satellite signal processing equipment has high reliability, and we requires special design to meet the above special application scenarios[4].

As a fast and accurate method of transcendental function operation, CORDIC algorithm has great accuracy and real-time[5]. CORDIC is widely used in real-time speech signal processing, image signal processing, communication engineering, detection and other applications. However, due to the rapid development of computer science, now researchers are more accustomed to using traditional computing devices such as video card, DSP / SOC architecture to solve and process data, but obviously this is not suitable for the special field of on-board signal processing. CORDIC algorithm can calculate many complex mathematical functions without floating-point operation,its implementation process only uses adder and shift register, which is very suitable for FPGA design and implementation, and very suitable for the system with strict constraints on hardware and software resources.

The application of CORDIC Algorithm in the field of signal processing has been researched by many people. For example, in reference[6-7], jumping is tried to reduce the times of CORDIC generation selection. The author of reference[8] studies the hyperbolic rotation of CORDIC algorithm and solves the application of hyperbolic sine cosine function. In reference[9], the author uses CORDIC to solve the calculation problem of exponential function and logarithmic function, and based on CORDIC algorithm, studies and designs a set of Based on the hyperbolic sine cosine solution e index as the base of exponential operation and logarithmic operation. However, at present, the application of CORDIC by researchers is limited to the solution of a specific formula. At present, there is no literature for in-depth study and summary of CORDIC's usage. Based on the author's experience in research and development of aerospace signal processing products in recent years, this paper first deduces several CORDIC iteration models and generation selection models, and then summarizes the main calculation methods that can be used by these CORDIC algorithms A set of universal CORDIC soft core is designed and implemented by mathematical formula to realize the above calculation functions. Finally, taking a certain type of application as an example, the actual implementation process of CORDIC algorithm on FPGA is introduced.

2. Basic principle of CORDIC algorithm

2.1. Iterative model

CORDIC algorithm was proposed by Voder in 1959[10]. In 1971, Walther unified the form of this algorithm. [11] Vails realized the CORDIC algorithm by FPGA for the first time[12]. CORDIC algorithm is a general generation algorithm. It realizes various complex mathematical function evaluation operations through the rotation and orientation operation of control vector in linear coordinate system, circular coordinate system and hyperbolic coordinate system[13]. Its basic idea is to rotate the vector continuously from the starting point according to a certain phase step, and then approach the final point gradually. Take the selection of circular model as an example, its starting point is (x_0, y_0), after a thousand times of loss rotation, it approaches the end point (x_n, y_n), and its vector rotation diagram is shown in Figure 1[14].

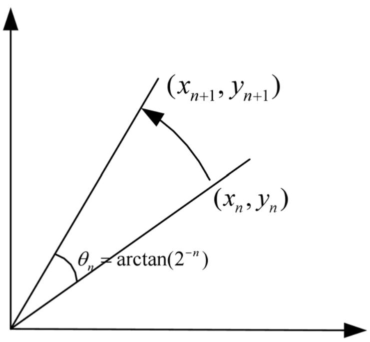

Fig. 1. Vector rotation for CORDIC

According to the coordinate iteration formula, we can get:

$$\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = \begin{bmatrix} \cos\theta_n & -\sin\theta_n \\ \sin\theta_n & \cos\theta_n \end{bmatrix} \begin{bmatrix} x_n \\ y_n \end{bmatrix} \tag{1}$$

Put $\cos\theta_n$ Forward and we'll get:

$$\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = \begin{bmatrix} 1 & -\tan\theta_n \\ \tan\theta_n & 1 \end{bmatrix} \begin{bmatrix} x_n \\ y_n \end{bmatrix} \cdot \cos\theta_n \tag{2}$$

Then the result after n iterations is:

$$\begin{bmatrix} x_n \\ y_n \end{bmatrix} = \prod_{i=1}^{n} \begin{bmatrix} 1 & -\tan\theta_i \\ \tan\theta_i & 1 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \cdot \prod_{i=1}^{n} \cos\theta_i \tag{3}$$

In order to facilitate the hardware implementation and simplify the operation, the phase of each generation selection is set to $\theta_n = \arctan(2^{-n})$ so when the number of iterations n is known, the $\prod_{i=1}^{n}\cos\theta_i$ in the above formula is a constant, which can be filtered by multiplying a k value in the input.

There are more commonly used iterative models such as hyperbolic,linear generation selection and exponential iteration, which will not be introduced here.

2.2. Generation selection mode

There are two iterative modes of CORDIC: rotation mode and vector mode. The difference is that rotation mode takes the angle approaching 0 as the selection criterion, while vector mode takes y approaching 0 as the sending criterion.

Similarly, taking the circular generation selection model as an example, if the rotation mode is selected, assuming that the input parameters are $(x_0, y_0) = (1,0)$ and $Z = \theta$, then t the result after n iterations is:

$$\begin{bmatrix} x_n \\ y_n \end{bmatrix} = \prod_{i=1}^{n} \begin{bmatrix} 1 & -\tan\theta_i \\ \tan\theta_i & 1 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \cdot \mathrm{K} \tag{4}$$

That is to say, it is equivalent to rotating (x_0, y_0) to θ in the coordinate system, and $(x_n, y_n) = (\cos\theta, \sin\theta)$ after rotation. Therefore, the selection of circle rotation can be used to calculate trigonometric functions. In actual operation, due to the existence of the constant term K, the input parameter $(x_0, y_0) = (K, 0)$ is usually set, and $K = \frac{1}{\prod_{i=1}^{n}\cos\theta_i}$.

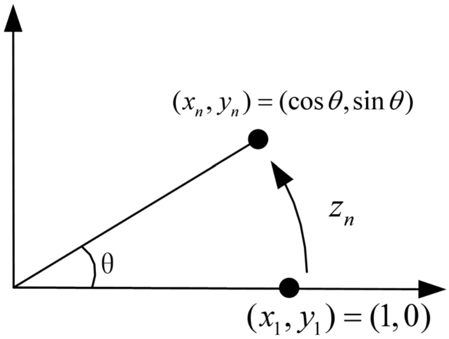

Fig. 2. Rotation mode for CORDIC

If the vector mode is selected, assuming that the input parameters are $(x_0, y_0) = (A, B)$ and $Z = 0$, then the generation result is equivalent to rotating (x_0, y_0) to the X-axis, where $(x_n, y_n) = (\sqrt{A^2 + B^2}, 0)$, $z_n = \arctan(B / A)$. This alternative mode is often used to calculate vector modules or inverse trigonometric functions.

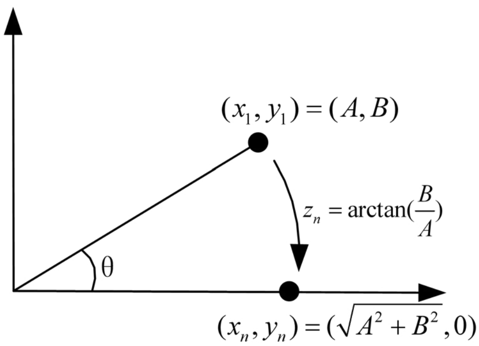

Fig. 3. Vector mode for CORDIC

Similarly, hyperbolic iteration model, linear iteration model and exponential iteration can also get different iteration results by rotating different iteration modes. No more detailed derivation here.

2.3. Summary

The subtlety of CORDIC algorithm is that the output precision of generation selection mode only depends on the number of iterations, and different generation selection modes only need to bind different generation selection parameters[15]. Therefore, CORDIC algorithm can be used to calculate a variety of complex mathematical functions. Here are the CORDIC solution methods of several commonly used mathematical functions that the author has used and implemented with FPGA in recent years, which can satisfy most needs of signal processing application scenarios.

Table 1. Calculate a variety for complex mathematical functions.

Iterative mode	Xin	Yin	Zin	Xout	Yout	Zout
Circumferential rotation	K	0	θ	Cosθ	Sinθ	0
Circumferential vector	K*A	K*B	0	$\sqrt{A^2+B^2}$	0	Atan(B/A)
Hyperbolic rotation	K	0	θ	Coshθ	Sinhθ	/
Hyperbolic rotation	K	K	θ	e^{θ}	/	/
Hyperbolic vector	K*A	K*B	0	$\sqrt{A^2-B^2}$	0	/
Hyperbolic vector	K*A+K/4	K*A-K/4	0	$\sqrt{A}$	0	/
Hyperbolic vector	K*A+K	K*A-K	0	/	0	Ln^A
Linear rotation	A	B	0	/	0	B/A
Exponential rotation	1	0	A	10^A	/	0
Exponential vector	A	0	0	0	/	Lg^A

Note: the correction value K shall be calculated in advance according to the number of iteration steps and iteration mode.

3. Engineering implementation of CORDIC algorithm

3.1. Basic operation architecture unit

The algorithm architecture only uses addition operation and shift operation[16], and has the same algorithm structure and different generation selection methods Only different iteration parameters (T) need to be bound.

The core computing architecture unit is shown in the figure below:

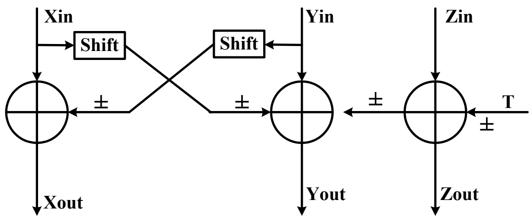

Fig. 4. The core computing architecture for CORDIC

If the real-time requirement of the application scenario is low, only the output of the above structure needs to be led back to the input,iIt can realize repeated generation calculation.

If the applied signal processing system requires high real-time performance, it can be realized by pipeline architecture. Input parameters are pipelined into CORDIC core, which can be obtained in each clock cycle.

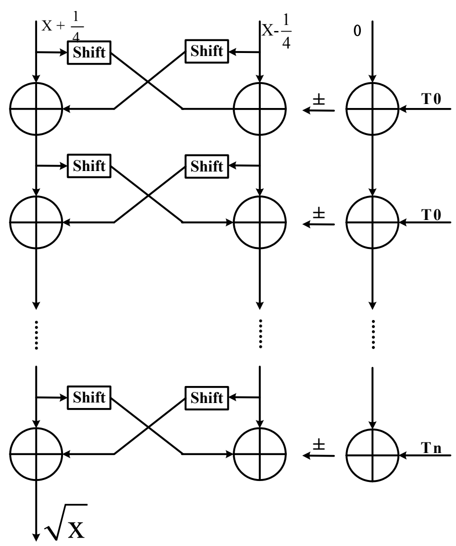

Fig. 5. Pipeline architecture for CORDIC

3.2. FPGA code implementation

FPGA code is implemented by VHDL code, and the whole CORDIC core is composed of preprocessing unit and core operation unit.

The task of preprocessing unit is to generate the initial value of core operation unit.

The processing steps are as follows:

- Read the iteration code and select the iteration type;
- The X andY value is multiplied by the corresponding correction factor K value;
- Add a constant value to X and Y, and complete the conversion from original code to complement code;
- Z value is converted to the value required by the core algorithm processing unit by angle transformation;

The functional block diagram of pretreatment unit is as follows:

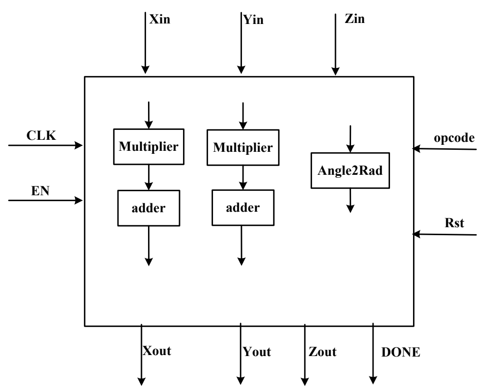

Fig. 6. Processing block diagram of pretreatment unit.

Similarly, take hyperbolic vector calculate Square as an example, and the corresponding relationship between input and output is as follows:

Table 2. Input and output of preprocessing unit.

Iterative mode	Xin	Yin	Zin	Xout	Yout	Zout	Opocode
Hyperbolic vector（Seeking the square root）	A	A	0	K*A+K/4	K*A-K/4	0	1101

The core operation unit is the core component of CORDIC core, and the process of this module is as follows:
Read selected code and select iteration type;

- Binding rotation factor according to control code;
- Import initial Y, Y, Z values;
- Generation computation;
- Output result;

The overall design module diagram of the core operation unit is shown in the figure below, and the state machine implementation is adopted for the generation of non flowing water. Now, the generation selection is defined as i. when the number of generation selection is less than or equal to 17, the state machine will switch down and the number of generation selection will be

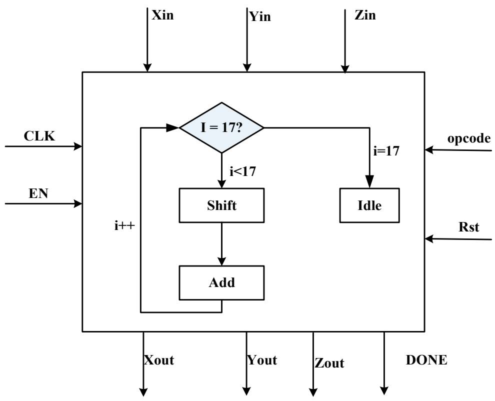

Fig. 7. Processing block diagram of core operation unit.

Similarly, take hyperbolic vector calculation as an example, and the corresponding relationship between input and output is as follows

Table 3. Input and output of preprocessing unit.

Iterative mode	Xin	Yin	Zin	Xout	Yout	Zout	Xin
Hyperbolic vector（Seeking the square root）	K*A+K/4	K*A-K/4	0	$\sqrt{A}$	0	/	K*A+K/4

3.3. Summary

In this chapter, FPGA/VHDL completes the hardware implementation of CORDIC core and the self-developed CORDIC core after encapsulation[17]. The iterative calculation of different mathematical functions can be completed by inputting different operation codes and loading different rotation factors. Compared with the core of Xilinx, the self-developed CORDIC core is far stronger in function and performance. At present, the fixed soft core structure can complete the calculation of the mathematical functions listed in table 1, and can cover most of application scenarios in the field of signal processing[18].

4. Application examples

4.1. Requirements overview

In order to meet the requirements of high-speed inter satellite communication and measurement and control, the inter satellite link subsystem has become the standard configuration on the satellite[19-20]. The inter satellite link antenna is an important link to establish the inter satellite link. In order to meet the needs of large bandwidth and rapid response, the Ka band phased array antenna is usually used, and the equipment is required to have long life, high reliability, high-speed rate, anti-interference and other important characteristics. At the same time, due to the high-speed movement of communication platform in space, phased array antenna needs to have a very high beam switching speed, so it is very difficult to design.

In order to meet the above functional requirements, the research group proposed a tile architecture system based on vector modulation and centralized wave control after detailed demonstration, as shown in the following figure:

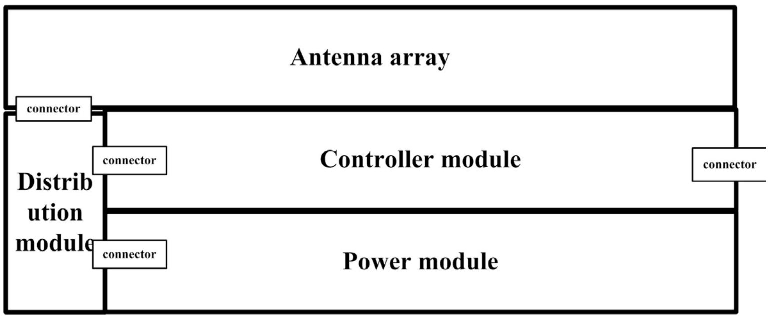

Fig. 8. Overall framework of Space-borne array antenna

4.2. Hardware design

Among them, the antenna array is realized by M (vector modulation). The phased array antenna of MM system has a very high ability to control the amplitude and phase. The disadvantage is that each RF unit needs to calculate the amplitude and phase values in real time according to the beam direction, and the amplitude and phase values of RF units will drift with time. Therefore, in order to ensure the stability of the wave control direction, it is necessary to carry out amplitude and phase correction one by one for RF units on a regular basis

The wave controller is the core part of the whole phased array antenna. If the conventional design method is used, at least one timing refresh FPGA + one ram FPGA + one DSP and related peripheral circuits are needed to realize the design. In order to meet the reliability requirements, AB machine design is also needed for the above circuits, which will lead to the wave controller's too large volume and weight to meet the refined and miniaturized design

For this reason, the research group adopts the refined design idea, the whole wave controller only includes a single circuit board, AB machine is distributed on the single board, and the wave control function and amplitude and phase correction function are realized by a single anti fuse FPGA. The anti fuse FPGA has the characteristics of single particle resistance, high reliability, and no external chip configuration. The wave controller architecture is shown in the figure below.

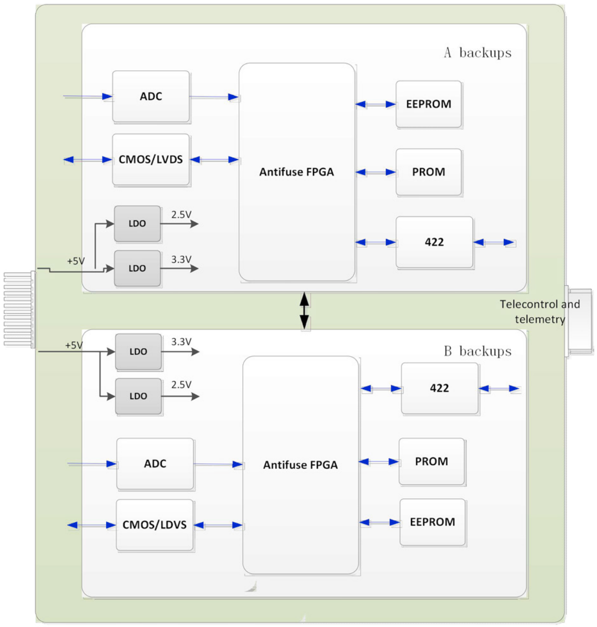

Fig. 9. Overall framework of the Controller

4.3. Implementation of CORDIC algorithm

The scale of antifuse FPGA is much smaller than RAM FPGA, and the commonly used antifuse FPGA in aerospace field is onlyTens of thousands of gates to hundreds of thousands of gates. In order to complete the high-speed wave control function under the condition of limited resources, the wave control function uses flow The CORDIC core of the pipeline architecture can output one RF channel in each clock cycle. The VM modulator of each TR component is configured quickly to control the direction switching of array beam. Wave control process is as follows:

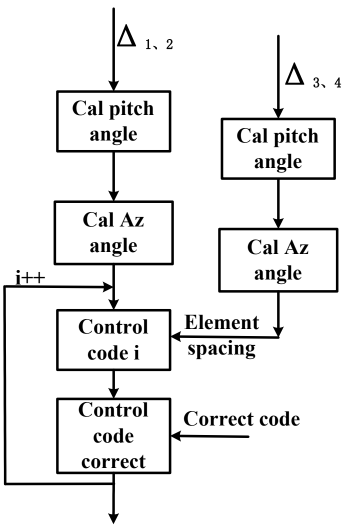

Fig. 10. Technological process of Wave control

Rotation vector method is a method to detect the amplitude and phase errors of each element of phased array, which is applicable to both the transmitting array and the receiving array. Error! Reference source not found. This method can calculate the amplitude and phase of the antenna by changing the value of the phase shifter of the unit to be corrected in turn and measuring the change value of the amplitude of the corresponding electric field vector, and store the correction result in EEPROM for wave control

Because the amplitude and phase correction function has no special requirements for real-time performance, the state machine + ordic iterative calculation architecture is used to complete. The state machine is responsible for the control algorithm flow. The corresponding mathematical operation is completed by inputting different operation codes to the CORDIC core in each step. The processing block diagram is as follows:

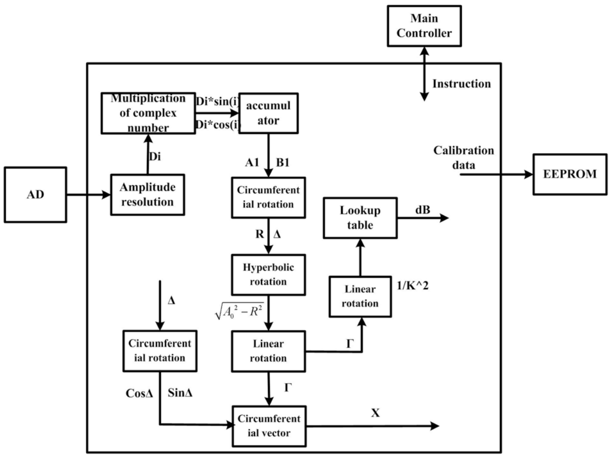

Fig. 11. Overall framework of Rotation vector

5. Conclusion

In orbit high-speed signal processing technology will be the main development direction of space borne system in the future. In view of the limitation of computing speed and resource in the field of satellite applications, a real-time signal processing architecture based on CORDIC algorithm is proposed and developed. On the basis of the same algorithm precision, the operation resource, operation speed and hardware requirements of the algorithm architecture are better than other design methods. The author's department will continue to carry out the research of system design and real-time signal processing application in the field of microwave remote sensing and SAR, and combine the onboard signal processing and CORDIC algorithm organically to realize the autonomous and intelligent operation of satellite system.

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